

Semantic Retrieval Infrastructure

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Popular Vector Database Platforms

Common vector databases include: - ChromaDB - Pinecone - Weaviate - Milvus - FAISS Each platform offers different tradeoffs around persistence, scalability, infrastructure complexity, metadata filtering, deployment models, and hybrid retrieval support. ChromaDB is commonly used for local RAG development because of its simplicity and strong integration with LangChain ecosystems.

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Retrieval Optimization

Enterprise retrieval systems optimize performance using: - hybrid retrieval - metadata filtering - semantic reranking - chunk overlap - adaptive retrieval pipelines Similarity search commonly uses cosine similarity, nearest-neighbor search, or approximate nearest-neighbor indexing algorithms. Retrieval quality depends heavily on: - embedding quality - chunk granularity - semantic overlap - reranking strategies - document preprocessing

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Operational Challenges

Large-scale enterprise deployments face operational challenges including: - indexing latency - embedding drift - storage growth - retrieval latency - infrastructure cost management Monitoring frameworks are essential for measuring: - retrieval precision - recall - hallucination rates - semantic relevance - workflow reliability Enterprise AI teams increasingly invest in retrieval evaluation pipelines to improve production system quality.

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